

Jooneui Hong

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Research Interests

- Transportation equity, Congestion pricing, Travel demand-supply modeling, Traffic data quality

Education

- **University of California, Irvine, CA, USA** 2022 - Present
Ph.D. student in Civil and Environmental Engineering
- **Seoul National University, Seoul, South Korea** 2016 – 2018
Master of Science in Civil and Environmental Engineering
Thesis: Implementing Surrogate Safety Measures to Driving Simulator and its Applicability
- **Korea University, Seoul, South Korea** 2010 – 2015
Bachelor of Science in Civil, Environmental, and Architectural Engineering
Sergeant, Army of the Republic of Korea (July. 2011 - July. 2013)
- **University of Leeds, Leeds, United Kingdom** Feb. 2015 – Jun. 2015
Exchange Student

Research Experience

- **Institute of Transportation Studies, UC Irvine** 2023 – Present
Research Assistant
 - Investigated carbon reduction strategies through innovative road pricing mechanisms, emphasizing transportation policies' role in achieving sustainability goals.
- **Public and Private Infrastructure Investment Management Center, Korea Development Institute (KDI)** 2018 – 2022
Research Associate
 - Conducted 8 government-funded Preliminary Feasibility Studies (PFS) for road and railway projects, ensuring efficient allocation of national funds under Korea's National Finance Act.
 - Constructed weekend trip origin-destination matrices using mobile data by defining conversion factors for weekday-to-weekend travel demand, improving travel demand accuracy.
 - Estimated traffic noise costs using a damage-cost approach, providing insights for cost-benefit analysis.
 - Developed macro-level traffic noise prediction models by scaling micro-noise models for travel demand analysis.
 - Co-authored and revised the 6th edition of the PFS guidelines for road and railway projects, standardizing evaluation methodologies.
 - Collaborated with key government ministries and research institutions, streamlining communication and ensuring cohesive project evaluations.
- **Department of Civil and Environmental Engineering, Seoul National University** 2016 – 2018
Research Assistant
 - Developed a driving simulator-based training program to reduce aggressive driving behaviors, leveraging advanced traffic flow models (car-following, lane-changing, gap-acceptance).
 - Integrated Surrogate Safety Measures (SSMs) into simulations to evaluate driver safety in interactive traffic environments.
 - Built a comprehensive driver scoring system using statistical methods, including factor analysis and principal component analysis (PCA), implemented in R.
 - Coordinated interdisciplinary collaborations with the Korea Transport Institute and the Korea Transportation Safety Authority to ensure the program's success.

Teaching & Mentoring Experience

- Teaching Assistant Jan. 2024 - Present
Department of Civil and Environmental Engineering, UC Irvine
– CEE122/229A: Operations & Control (Winter quarter 2024, Winter quarter 2025)
- Mentorship: Alex Wang, Development of week-to-week stationary states traffic state method. *Emerging Scholars Transportation Research Showcase (PRIME/TRIP)*.
- Mentorship: Alex Wang, Development of week-to-week stationary states traffic state method. *Emerging Scholars Transportation Research Showcase (PRIME/TRIP)*, Institute of Transportation Studies, University of California, Irvine, CA, USA, Jun. – Sep. 2025.
- Mentorship: Charlie Zhao, Data analysis using the large-scale video image processing data (I-24 MOTION data) in Tennessee. *Emerging Scholars Transportation Research Showcase (PRIME/TRIP)*, Institute of Transportation Studies, University of California, Irvine, CA, USA, Jun. – Sep. 2024.

Publications

- Lee, J., Moh, D., **Hong, J.**, & Lee, C. (2021). The effect analysis of safe driving education for high-risk driver groups in sudden pedestrian crossing situation using a driving simulator. *The Journal of The Korea Institute of Intelligent Transport Systems*. 2021;20(5):18-34. [Link]
- Ka, E., Kim, D., **Hong, J.**, & Lee, C. (2020). Implementing Surrogate Safety Measures to Driving Simulator and Evaluating the Safety Effects of Simulator-Based Training on Aggressive Driving Behaviors. *Journal of Advanced Transportation*, 2020(1), 7525721. [Link]

Refereed Conference Proceedings

- **Hong, J.**, & Jin, W. (2026). Data-driven peak-period detection for improved volume-delay function calibration. In *Proceedings of the 4th Irvine Symposium on Emerging Research in Transportation (ISERT 2026)*, University of California, Irvine, CA, USA.
- **Hong, J.**, & Jin, W. (2025). Detecting traffic sensor malfunctions through lane-to-lane correlation analysis: A comparative study using NGSIM and PeMS datasets. In *Proceedings of the 2025 PSR Annual Congress*, University of California, Berkeley, CA, USA.
- **Hong, J.**, & Jin, W. (2025). Detecting traffic sensor malfunctions through lane-to-lane correlation analysis: A comparative study using NGSIM and PeMS datasets. In *Proceedings of the SoCal CEE Research Symposium (SoCal CEERS 2025)*, University of California, Irvine, CA, USA.
- **Hong, J.**, & Jin, W. (2025). Detecting traffic sensor malfunctions through lane-to-lane correlation analysis: A comparative study using NGSIM and PeMS datasets. *Transportation Research Board 104th Annual Meeting*, Washington, DC, USA.
- **Hong, J.**, Kim, S., Ka, E., & Lee, C. (2017). Application of support vector machine (SVM) to judge the risky driver in driving simulator. *Proceedings of the 2017 Fall Korea Institute of ITS Conference*, Seoul, South Korea.
- Ka, E., Ha, S., **Hong, J.**, & Lee, C. (2017). Application of deep learning for demand forecasting of call taxi for the handicapped in Seoul. In *The 77th Korean Society of Transportation Conference*, Seoul, South Korea.
- **Hong, J.**, Ka, E., Kim, S., & Lee, C. (2017). Analysis and comparison of drivers between South Korea and the United States of America by using surrogate safety measures (SSMs). In *The 6th International Conference of Euro Asia Civil Engineering Forum*, Seoul, South Korea.
- Ka, E., Kim, S., **Hong, J.**, & Lee, C. (2017). A preliminary study of comparison with lane changing model parameters in merging area between normal and raining conditions. In *The 12th International Conference of Eastern Asia's Society for Transportation Studies (EASTS)*, Ho Chi Minh, Vietnam.
- **Hong, J.**, Lee, S., & Lee, C. (2016). Drivers' evaluation indices considering the interaction with surrounding vehicles on driving simulator. In *The 3rd International Conference on Computational Science and Engineering (ICST)*, Ho Chi Minh City, Vietnam.

- **Hong, J., Kim, J., Lee, J., & Lee, C. (2016).** A statistical method to assess overall driving risk using driving simulator. In *Proceedings of the 2016 Fall Korea Institute of ITS Conference*, Busan, South Korea.
- **Hong, J., Ka, E., Kim, S., & Lee, C. (2016).** Selecting a variety of surrogate safety measures (SSMs) using driving simulator. In *The 75th Korean Society of Transportation Conference*, Busan, South Korea.

Awards & Honors

- **2026 KOTAA Student Service Award** Jan. 2026
Korean Transportation Association in America (KOTAA)
- **2025 KOTAA Student Service Award** Jan. 2025
Korean Transportation Association in America (KOTAA)
- **2025 KOTAA Travel Grant Award** Jan. 2025
Korean Transportation Association in America (KOTAA)
- **UCLA Bruin AI Responsibility Summit - Most Innovative (1st place out of 18 teams)** Jun. 2024
Developed and tuned machine learning models for vehicle trajectory data smoothing, utilizing architectures including Denoising Autoencoder Neural Network, Random Forest, and Gradient Boosting
- **2024 KOTAA Student Service Award** Jan. 2024
Korean Transportation Association in America (KOTAA)
- **Excellent Student Service Award** Sep. 2017 - Feb. 2018
Seoul National University
Amount: \$4,200 (100% of tuition + \$1,000)
- **Academic Excellence Scholarship** Sept. 2017 - Feb. 2018
Seoul National University
Amount: \$4,200 (100% of tuition + \$1,000)
- **LOTTE Foundation Scholarship** Sep. 2014 - Feb. 2016
Amount: \$12,000 (100% of tuition for three semesters)
- **Engineering Future Talent Scholarship** Mar. 2011 - Aug. 2011
Korea University, College of Engineering
Amount: \$2,000 (50% of tuition)

Extracurricular Activities

- **Exchange Student at the University of Leeds** Mar. 2015 - July 2015
 - Determined to study transportation engineering.
 - Participated in various societies, including Fishing, Football, Photography, and Frisbee.
- **Korea University Buddy Assistants (KUBA)** Feb. 2015 - Jun. 2015
 - Assisted exchange students in adapting to Korean culture and campus life.
 - Organized and participated in events such as welcoming parties, weekly dinners, monthly field trips, and international student festivals.
 - Facilitated language exchange programs by teaching Korean and improving English proficiency.
- **Captain, Department of Engineering Soccer Club** Feb. 2014 - Dec. 2014
 - Led the Department of Engineering soccer club to two Korea University league championships.
- **Sergeant, Army of the Republic of Korea** July 2011 - July 2013
 - Served at the Special Duty Team in South Korea's Air Force.

Professional Training

- **Statistical Analysis with R Programming**

Seoul National University, Seoul, Korea

July 2017

- Focused on implementing multivariate statistics (Clustering, and Principle component analysis, etc.) and classification (Logistic regression, etc.) using R.

Skills

- Programming: R, MATLAB, Python, C, Fortran
- OS: Linux
- Software: Emme4, VISSIM, CPLEX, SPSS, MS Excel, MS Access